

Towhid Ahmed

LLM Engineer

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Professional Summary

LLM Engineer with **1+ year of freelance experience** building **production-grade LLM applications, multi-agent systems, and RAG** solutions for SaaS and e-commerce platforms. Developed 3 AI applications using **LangChain, LangGraph, and FastAPI**, achieving **35% hallucination reduction, 25% retrieval relevance improvement, and 98/98 passing automated tests**. Experienced in **RAG pipelines, LLMops, Docker, and LangSmith**.

Technical Skills

Programming: Python, SQL

LLMs & Generative AI: LLMs, Llama, OpenAI API, Groq, Prompt Engineering, AI Agents

RAG & Retrieval: RAG, Advanced RAG, Semantic Search, Vector Search, Cross-Encoder Re-ranking, Pinecone

AI Frameworks: LangChain, LangGraph, Hugging Face Transformers, Sentence Transformers

LLMops & Observability: LangSmith, LLM Evaluation, LLM Guardrails

Backend & Deployment: FastAPI, REST APIs, Async Programming, Docker

Testing & Software Engineering: OOP, Data Structures & Algorithms (Solved 400+ problems), LLM System Design

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | Jun 2026 – Jul 2026

- Enhanced information reliability and user trust by developing a **3-agent LangGraph workflow** with **citation-based RAG**, significantly **reducing AI hallucinations**.
- Powered seamless document processing and interaction by engineering **7 production-ready REST APIs** using **FastAPI, OpenAI API, and Pinecone**.
- Enabled **1,000+** document knowledge-base updates without redeployment.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- **Automated customer interactions and improved consistency** using a multi-agent **LangGraph pipeline** (Intent, Retrieval, Response) with Pinecone RAG and **conversational memory**.
- Slashed AI inference costs and maintained high execution speed by integrating **Groq**, Hugging Face embeddings, and **Pinecone Serverless**.
- **Built a scalable support infrastructure** with a Streamlit UI and a FastAPI backend featuring **6+ REST APIs** for knowledge uploads and session management.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, Hugging Face, Tavily GitHub | Demo

- **Designed 4-agent LangGraph workflow** routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily** with **DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25% and reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker**; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built a clinical information assistant using a **4-agent LangGraph pipeline** (Router, Safety, Retrieval, Answer) with **RAG and web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved **98/98** passing tests across **guardrails, RAG evaluation, and API workflows**.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology | 2023-2027

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward